

§ 3.6 Moduli spaces and Quotients.

Last time, we defined good quotients and categorical quotients.

They help us to construct moduli spaces.

Recall:

(Recall from 03. Chi Ki's notes)

Def: (local universal property)

$$\begin{aligned} \mathcal{M}(S) &= \{ \mathcal{F} \text{ over } S \} / \sim_S \\ \mathcal{M}(\mathcal{F}: T \rightarrow S) &= \mathcal{F}^*: \mathcal{M}(S) \rightarrow \mathcal{M}(T). \end{aligned}$$

For a moduli problem $\mathcal{M}$ (i.e. a moduli functor $\mathcal{M} \in \text{Psh}(\text{Sch})$),
a family $\mathcal{F}$ over a scheme S has the local universal property if
 $\forall$ family $\mathcal{G}$ over T and $\forall$ k -point $t \in T$,
 $\exists$ a nbh $U \ni t$ in T and morphism $f: U \rightarrow S$ s.t. $\mathcal{G}|_U \sim_U f^* \mathcal{F}$.

Proposition: (3.35 in lecture notes)

For a moduli problem $\mathcal{M}$,

let $\mathcal{F}$ be a family over S with local-universal property.

Suppose $\exists$ algebraic group G acting on S , s.t.

$\forall$ two k -points $s, t \in S(k)$, $G \cdot s = G \cdot t$ iff $\mathcal{F}_s \sim \mathcal{F}_t$

Then

↑
Orbit

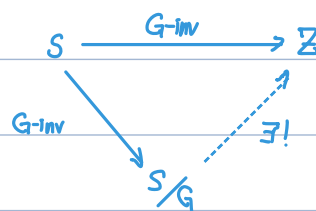
↑
fiber over s

(a) Any coarse moduli space is a categorical quotient of $G \curvearrowright S$.

Recall: $(\mathcal{M}, \eta)$, $\eta: \mathcal{M} \rightarrow h_{\mathcal{M}}$ s.t.

① $\eta_{\text{Spec } k}: \mathcal{M}(\text{Spec } k) \rightarrow h_{\mathcal{M}}(\text{Spec } k)$ is bijective

② $\forall$ scheme N and $v: \mathcal{M} \rightarrow h_N$, $\exists! f: \mathcal{M} \rightarrow N$ s.t. $v = h_f \circ \eta$.



(b) A categorical quotient of $G \curvearrowright S$ is a coarse moduli space
iff it is an orbit space.

↑
i.e. the preimage of each k -point is a single orbit.

Idea of proof:

① Show bijection: $\forall$ scheme M ,

$$\{\text{natural transformation } \eta: \mathcal{M} \rightarrow h_M\} \longleftrightarrow \{G\text{-invariant morphisms } f: S \rightarrow M\}.$$

$$\eta \longmapsto \eta_S(\mathcal{F})$$

$$\mathcal{M}(S) \rightarrow \text{Hom}(S, M)$$

② (Proving (a))

If (M, η) is a coarse moduli space,

$$\begin{array}{ccc} S \xrightarrow{\varepsilon_S(\mathcal{F}), G\text{-inv}} \mathbb{Z} &] & \text{corresponds to } \mathcal{M} \xrightarrow{\varepsilon} h_{\mathbb{Z}} \\ \eta_S(\mathcal{F}), G\text{-inv} \searrow & \nearrow \exists! f & \downarrow \text{(By def of coarse Ms)} \\ M & \Leftarrow \exists! f: M \rightarrow \mathbb{Z}, \text{ s.t. } \varepsilon = h_f \circ \eta & \end{array}$$

$$(h_f \circ \eta)_S(\mathcal{F}) = h_{f,S} \circ \eta_S(\mathcal{F})$$

$$= f \circ \eta_S(\mathcal{F})$$

$$\left(\begin{array}{l} f: M \rightarrow \mathbb{Z} \\ h_f: \text{hom}(-, M) \xrightarrow{f^*} \text{hom}(-, \mathbb{Z}) \end{array} \right)$$

② (Proving (b))

$\eta_S(\mathcal{F}): S \rightarrow M$ is orbit space

iff preimage of each k -point is a single orbit.

iff $\eta_{\text{Spec } k}: \underbrace{\mathcal{M}(\text{Spec } k)}_{\substack{\text{families over } k / \sim \\ \parallel \\ \text{orbits}}} \rightarrow \underbrace{\text{hom}(\text{Spec } k, M)}_{k\text{-points of } M}$ is bijective.

§4 Affine geometric invariant theory.

In this chapter, G : affine algebraic group

X : affine scheme of f.t. $/\mathbb{k}$.

($\mathcal{O}_X(X)$ is f.g. $\mathbb{k}$ -algebra.)

We will show $G \curvearrowright X$ has a good quotient when G is linearly reductive.

Recall: Group action on scheme induces ring map in the opposite direction.

$$\begin{aligned}\sigma: G \times X &\longrightarrow X \\ (g, x) &\longmapsto g \cdot x\end{aligned}$$

$$\leadsto \sigma^*: \mathcal{O}(X) \longrightarrow \mathcal{O}(G \times X) = \mathcal{O}(G) \otimes_{\mathbb{k}} \mathcal{O}(X).$$

$$f \longmapsto [(g, x) \mapsto f(g^{-1} \cdot x)]$$

$$= \sum h_i \otimes f_i$$

depends on the action.

$$\leadsto \text{linear representation } G \rightarrow GL(\mathcal{O}(X))$$

$$g \longmapsto [f \mapsto \sum h_i(g) \otimes f_i]$$

$$\parallel \\ [f(x) \mapsto f(g^{-1} \cdot x)]$$

The ring of G -invariant regular functions on X is

$$\mathcal{O}(X)^G := \{f \in \mathcal{O}(X) : \sigma^*(f) = 1 \otimes f\}$$

$$= \{f \in \mathcal{O}(X) : f(g \cdot x) = f(x) \quad \forall g \in G\}.$$

If $\varphi: X \rightarrow Z$ is a G -invar. morphism of schemes, then

$\varphi^*: \mathcal{O}(Z) \rightarrow \mathcal{O}(X)$ has image contained in $\mathcal{O}(X)^G$.

$$(f \mapsto f(\varphi(x)), \text{ but } g \cdot f(\varphi(x)) = f(\varphi(g^{-1} \cdot x)) = f(\varphi(x)).)$$

§4.1 Hilbert's 14th problem.

Recall def:

A : k -algebra,

$G \curvearrowright A$ is rational if every elt. of A is contained in a finite dim G -invar linear subspace of A .

Question: (Hilbert)

G : affine alg. group, A : f.g. k -algebra.

If $G \curvearrowright A$ is rational, is the algebra of G -invariants A^G finitely generated?

Answer (Nagata): Not in general,

Yes if G is a reductive group.

We will only prove the case when G is linearly reductive.

§4.2 Reductive groups

↳ 3 notions: reductive, linearly reductive, geometrically reductive

First goal: Jordan decomposition for affine algebraic groups over k .

Recall: $\forall g \in GL_n(k)$,

$$g = g_{ss} g_u = g_u g_{ss},$$

g_{ss} : semisimple (= diagonalizable since $k = \bar{k}$)

g_u : unipotent (i.e. $g - I_n$ is nilpotent.)

• For arbitrary affine algebraic group G , want analogous definition using the fact that G has a faithful linear rep $G \hookrightarrow GL_n$.

• Also, want the decomposition $g \mapsto (g_{ss}, g_u)$ to be functorial wrt closed immersions of groups, i.e.

$f: G \rightarrow H$ map of aff. alg. groups, then $\forall g \in G$, $f(g_{ss}) = f(g)_{ss}$, $f(g_u) = f(g)_u$.

Definition 4.1 : (Semisimple, unipotent)

Let G be an affine algebraic group / k .

An elt $g \in G$ is semisimple if there is a faithful linear rep

$\rho: G \hookrightarrow GL_n$ s.t. $\rho(g)$ is diagonalizable.
unipotent.

Theorem 4.2 (Jordan decomposition)

Let G be an affine algebraic group / k .

$\forall g \in G(k)$ (k -points), $\exists!$ semisimple g_{ss} , $!$ unipotent g_u , s.t.

$$g = g_{ss} g_u = g_u g_{ss}.$$

Moreover, this decomp is functorial w.r.t. group homo.

In particular, $\forall \rho: G \rightarrow GL_n$, ρ preserves semisimpleness & unipotency.

Def: (G -invariant vector / vector subsp.)

Let $\rho: G \rightarrow GL(V)$,

$$\begin{aligned} \rho^*: V &\rightarrow \mathcal{O}(G) \otimes_k V \\ v &\mapsto \underline{\sum h_i \otimes v_i} = \rho(\cdot^{-1}) \cdot v \end{aligned}$$

$V' \subset V$ is G -inv if $\rho^*(V') \subseteq \mathcal{O}(G) \otimes V$
 $(\rho(g)v \in V' \ \forall v \in V', g \in G)$

$v \in V$ is G -inv if $\rho^*(v) = 1 \otimes v$.
 $(\rho(g)v = v \ \forall g \in G)$

Def 4.3 : (unipotent group)

An aff. alg. group G is unipotent if

every non-trivial linear rep. $\rho: G \rightarrow GL(V)$ has a nonzero G -inv. vector.

Proposition 4.4 : (Eqv. def. of unipotent groups.)

TFAE :

(i) G is unipotent.

(ii) $\forall \rho: G \rightarrow GL(V)$, $\exists$ basis of V s.t. $\rho(G)$ is contained in the subgroup $U = \{\text{upper triangular matrices with diag. entry } = 1\} \subseteq GL(V)$

(iii) G is isom. to a subgroup of a standard unipotent group $U_n = \dots$ of GL_n .

(Proof skipped)

Rk: G is unipotent $\Rightarrow$ all ets are unipotent.

$\Leftarrow$ requires smooth.

Example of unipotent groups

eg. The additive group G_a .

$$G_a \hookrightarrow U_2 \text{ by } c \longmapsto \begin{bmatrix} 1 & c \\ 0 & 1 \end{bmatrix}$$

Def 4.7 : (Normal subgroup)

$H \triangleleft G$ if the conjugation action

$$H \times G \longrightarrow G \quad \text{factors through } H \hookrightarrow G.$$
$$(h, g) \longmapsto ghg^{-1}$$

Def 4.8 : (Reductive)

G is reductive if it is smooth and

every smooth unipotent normal alg. subgroup of G is trivial.

(Ex 4.10) GL_n and SL_n are reductive.

Def 4.11 (Linearly reductive, Geometrically reductive)

- ① G is linearly reductive if every finite dim linear rep is completely reducible, i.e., decomposes as a direct sum of irreducibles.
- ② G is geometrically reductive if $\forall$ finite dim linear rep $\rho: G \rightarrow GL(V)$ and G -invar. $0 \neq v \in V$,
 $\exists$ non-const. homog. $f \in \mathcal{O}(V)^G$ s.t. $f(v) \neq 0$.

Example: Any algebraic torus $(G_m)^r$ is linearly reductive by

Prop 3.12

Proposition 3.12. For a finite dimensional linear representation of a torus $\rho: T \rightarrow GL(V)$, there is a weight space decomposition

$$V \cong \bigoplus_{\chi \in X^*(T)} V_\chi$$

where $V_\chi = \{v \in V : t \cdot v = \chi(t)v \forall t \in T\}$ are called the weight spaces and $\{\chi : V_\chi \neq 0\}$ are called the weights of the action.

Proposition 4.14 (Eqv. def of linearly reductive).

TFAE:

- (i) G is linearly reductive.
- (ii) $\forall$ finite dim lin. rep $\rho: G \rightarrow GL(V)$,
any G -inv. subspace $V' \subseteq V$ has a G -stable complement.
- (iii) $\forall$ surjection of finite dim linear G -rep $\phi: V \rightarrow W$,
the induced map on G -invariants $\phi^G: V^G \rightarrow W^G$ is surjective.
- (iv) $\forall$ finite dim lin. rep $\rho: G \rightarrow GL(V)$, $\forall 0 \neq v \in V^G$,
 $\exists$ G -invariant linear form $f: V \rightarrow k$ s.t. $f(v) \neq 0$.
- (v) $\forall$ finite dim lin. rep $\rho: G \rightarrow GL(V)$, $\forall$ surjective linear form $f: V \rightarrow k$,
 $\exists v \in V^G$ s.t. $f(v) \neq 0$.

Proof sketch :

(i) $\Leftrightarrow$ (ii) : clear .

(ii) $\Rightarrow$ (iii) : $f: V \twoheadrightarrow W$,

$V' := \ker(f) \subset V$ is G -invariant.

$\Rightarrow V = V' \oplus V''$, where $V'' \cong W$.

$$V^G = V'^G \oplus V''^G$$

$\Rightarrow f^G: V^G \rightarrow (V'')^G = W^G$ is surjective.

(iii) $\Rightarrow$ (ii) : Let $\rho: G \rightarrow GL(V)$, $V' \subset V$ G -invariant.

$\text{Hom}(V, V')$, $\text{Hom}(V', V')$ are G -reps: $g \cdot f \mapsto f(g-)$

Then we have surjection $\phi: \text{Hom}(V, V') \longrightarrow \text{Hom}(V', V')$ by restriction.
 $\uparrow$ this a map of G -reps.

$$\text{By (iii), } \phi^G: \text{Hom}(V, V')^G \twoheadrightarrow \text{Hom}(V', V')^G$$
$$f \longmapsto \text{id}_{V'}$$

Then $V = V' \oplus \ker f$.

(iv) $\Leftrightarrow$ (v) : clear by $V^G = \text{Hom}(\mathbb{k}, V)^G = \underline{\text{Hom}(V^\vee, \mathbb{k})}^G = (V^\vee)^\vee^G$

(iii) $\Rightarrow$ (iv) : finite dim lin. rep $\rho: G \rightarrow GL(V)$, $0 \neq v \in V^G$,

$$\text{Get } \phi: V^\vee \longrightarrow \mathbb{k} \leftarrow \text{trivial rep of } G.$$
$$f \longmapsto f(v)$$

$\Rightarrow \phi$ is a surjection of G -reps. By (iii), $\phi^G: (V^\vee)^G \twoheadrightarrow \mathbb{k}^G$.

$$f \longmapsto 1$$

such $f(v) = 1$.

(iv) $\Rightarrow$ (iii) : Let $\phi: V \twoheadrightarrow W$, want $\phi^G: V^G \twoheadrightarrow W^G$.

Let $w \in W^G$. By (iv), let $f: W \rightarrow \mathbb{k}$ G -inv st. $f(w) \neq 0$.

Then $f \circ \phi: V \rightarrow \mathbb{k}$ is G -inv, surjective.

By (v) $\Leftrightarrow$ (iv), $\exists v \in V^G$ st. $(f \circ \phi)(v) \neq 0 \Rightarrow (f \circ \phi)(v) = f(w) \Rightarrow \phi(v) = w$ after scaling v

Theorem 4.16 : (Weyl, Nagata, Mumford, Haboush ; Relating the 3 notions)

(i) linearly reductive $\Rightarrow$ geometrically reductive

(ii) In char 0, reductive $\Rightarrow$ linearly reductive.

(iii) If G is smooth, then G reductive $\Leftrightarrow G$ geometrically reductive.

Hence

G smooth : $\text{lin red} \Rightarrow \text{geom red} \Leftrightarrow \text{red}$

G smooth & char $k=0$: All equivalent.

Rk 4.17 :

For char $k > 0$, GL_n , SL_n , PGL_n are NOT lin. red for $n > 1$.

Exercise:

Exercise 4.10. Show that the general linear group GL_n and the special linear group SL_n are reductive. [Hint: if we have a non-trivial smooth connected unipotent normal algebraic subgroup $U \subset GL_n$, then there exists $g \in U(k) \subset GL_n(k)$ whose Jordan normal form has a $r \times r$ Jordan block for $r > 1$ (as g is unipotent). Using normality of U , find another element $g' \in U(k)$ such that the product gg' is not unipotent.]

WTS GL_n is reductive, i.e.

① smooth ✓

② Every smooth unipotent normal subgroup is trivial.

Suppose U is a unipotent normal subgroup.

Let $I_n \neq g \in U$

g has a nontriv. Jordan Block.

$$J = \begin{bmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{bmatrix}, \quad g = \begin{bmatrix} J & \\ & * \end{bmatrix}$$

Take $A = \begin{bmatrix} & 1 & \\ & & \\ & & I \end{bmatrix} \in GL_n$

Then $AgA^{-1} = \begin{bmatrix} J^T & \\ & * \end{bmatrix}$

$g, AgA^{-1} \in U$

$\Rightarrow g \cdot (AgA^{-1}) = \begin{bmatrix} J \cdot J^T & \\ & *^2 \end{bmatrix} \in U$

But $J \cdot J^T$ is not unipotent:

(eg. block size of $J=3$.)

$$J \cdot J^T = \begin{bmatrix} 1 & & \\ & 1 & \\ & & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & & \\ & 1 & \\ & & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

is not unipotent since $\text{tr}(J \cdot J^T) = 5 \neq 3$

Exercise 4.13. Show directly that the additive group $\mathbb{G}_a$ is not geometrically reductive. [Hint: there is a representation $\rho: \mathbb{G}_a \rightarrow \mathrm{GL}_2$ and a G -invariant point $v \in \mathbb{A}^2$ such that every non-constant G -invariant homogeneous polynomial in two variables vanishes at v].

Not geom. red: $\exists$ finite dim rep $G \rightarrow \mathrm{GL}(V)$, $\exists$ nonzero G -inv. $v \in V$ st.

$\forall$ G -inv. nonconstant homog. poly $f \in \mathcal{O}(V)$, $f(v) = 0$.

$$\begin{aligned} \rho: \mathbb{G}_a &\longrightarrow \mathrm{GL}_2 \\ c &\longmapsto \begin{bmatrix} 1 & c \\ 0 & 1 \end{bmatrix} \end{aligned}$$

$$v = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \text{ is a } G\text{-invariant point in } \mathbb{A}^2: \begin{bmatrix} 1 & c \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Let $f \in \mathcal{O}(V)^G$ non-constant homog.

$$\Rightarrow f\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = f\left(\begin{bmatrix} 1 & c \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}\right) = f\left(\begin{bmatrix} x+cy \\ y \end{bmatrix}\right)$$

Since c is arbitrary, f is independent of the 1st entry.

$$\Rightarrow f\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = ay^r \quad \forall f \in \mathcal{O}(V)^G$$

$$\Rightarrow f\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = 0 \quad \forall f \in \mathcal{O}(V)^G$$

$\therefore \mathbb{G}_a$ is not geom. red.